

Hussein Shalaby

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EDUCATION

University of Padua

Bachelor of Science in Information Engineering

Padua, Italy

Expected Feb. 2027

PROJECTS

Renderer | *C++*, *OpenGL*, *Vulkan* | [github](#)

- Built a renderer with model loading, PBR shading, image-based lighting, HDR, and bloom.
- Implemented 8x MSAA and real-time shadows for directional, spot, and point lights.
- Started a Vulkan backend alongside OpenGL and built a Dear ImGui interface for debugging renderer settings.

CUDA Edge Detection | *CUDA*, *C++* | [github](#)

- Implemented a Sobel edge detector in CUDA and verified its output against a CPU implementation.
- Used Nsight Compute to compare kernel implementations and found the shared-memory version was about 1.8x slower than the naive version.

3D Gaussian Splatting Renderer | *Rust*, *wgpu*, *rust-gpu* | [github](#)

- Built a real-time 3D Gaussian Splatting renderer in Rust using wgpu, handling around 200k Gaussian splats.
- Wrote GPU shaders in Rust using rust-gpu and handled back-to-front splat sorting on the CPU.

OPEN SOURCE

Filament (Google) | *C++* – [contribution](#)

- Added `to_string()` conversions for public enums in Google's Filament real-time rendering engine.

Mesa (zink) | *C*, *Vulkan* – [contribution](#)

- Updated zink to use the promoted Vulkan 1.4 names for line-rasterization APIs that previously used EXT/KHR suffixes.

ACHIEVEMENTS

ICPC SWERC 2024 Finalist | *Southwestern Europe Regional*

Pisa, Italy

- Represented the University of Padova at the ICPC SWERC 2024 regional contest, competing against 180+ teams from southwestern Europe.

Italian Collegiate Programming Contest (ITACPC)

Italy

- Placed in the top 20 nationally in 2025 and top 30 in 2024 among 200+ university teams.

MATE ROV Regional – 4th place | *Underwater Robotics*

Alexandria, Egypt

- Developed Python/OpenCV image-processing code for autonomous missions in an underwater ROV.

EXPERIENCE

Freelance Software Developer

Aug. 2025 – Sep. 2025

Self-Employed

Remote

- Reduced active-window detection latency from around 5 seconds to under 100ms by switching to native platform event hooks.
- Built a Python-based parametric CAD tool for automating pressure-vessel layouts, reducing hours of manual work for clients.

IT Systems Intern

Aug. 2023 – Sep. 2023

Intercom Enterprises

Cairo, Egypt

- Completed Dell ISMv5, Enterprise Cloud, and RHEL training focused on datacenter virtualization and storage.
- Reviewed rack storage configurations as part of an enterprise disaster-recovery assessment, with a focus on load balancing.

TECHNICAL SKILLS

Languages: C++, C, Rust, Python, Go, SQL

Development: Git, CMake, GDB, Docker, Linux

Graphics & GPU: OpenGL, Vulkan, CUDA, GLSL, HLSL, RenderDoc, Nsight